

ABEKA SCHOLAR ACADEMY

Subj Grade K God's World (science) Curriculum

Official Curriculum & Study Material

Chapter 1

Welcome to the comprehensive, massive study module for subj-grade-K-god's world (science). We have an enormous amount of highly detailed academic material to cover today. Prepare yourself. Science is a systematic discipline that builds and organises knowledge in the form of testable hypotheses and predictions about the universe. Modern science is typically divided into two – or three – major branches: the natural sciences, which study the physical world, and the social sciences, which study individuals and societies. While referred to as the formal sciences, the study of logic, mathematics, and theoretical computer science are typically regarded as separate because they rely on deductive reasoning instead of the scientific method as their main methodology. Meanwhile, applied sciences are disciplines that use scientific knowledge for practical purposes, such as engineering and medicine. The history of science spans the majority of the historical record, with the earliest identifiable predecessors to modern science dating to the Bronze Age in Egypt and Mesopotamia (c. 3000–1200 BCE). Their contributions to mathematics, astronomy, and medicine entered and shaped the Greek natural philosophy of classical antiquity and later medieval scholarship, whereby formal attempts were made to provide explanations of events in the physical world based on natural causes; while further advances, including the introduction of the Hindu–Arabic numeral system, were made during the Golden Age of India and Islamic Golden Age.

The recovery and assimilation of Greek works and Islamic inquiries into Western Europe during the Renaissance revived natural philosophy, which was later transformed by the Scientific Revolution that began in the 16th century as new ideas and discoveries departed from previous Greek conceptions and traditions. The scientific method soon played a greater role in the acquisition of knowledge, and in the 19th century, many of the institutional and professional features of science began to take shape, along with the changing of "natural philosophy" to "natural science". New knowledge in science is advanced by research from scientists who are motivated by curiosity about the world and a desire to solve problems. Contemporary scientific research is often highly collaborative and is frequently carried out by teams in academic and research institutions, government agencies, and companies. At the same time, many major advances—particularly in fundamental science—have come from individual researchers and are widely recognised through major international awards such as the Nobel Prize. The practical results of scientific work have led to the emergence of science policies that seek to prioritise the responsible development of commercial products, health care, public infrastructure, environmental protection, and defense capabilities. The word science has been used in English since

the 14th century (Middle English) in the sense of "the state of knowing". The word was borrowed from the Anglo-Norman language as the suffix *-cience*, which was borrowed from the Latin word *scientia*, meaning 'knowledge, awareness, understanding', a noun derivative of *sciens* meaning 'knowing', itself the present active participle of *sci* Ò wFò ¶æðw'.

There are many hypotheses for science's ultimate word origin. According to Michiel de Vaan, Dutch linguist and Indo-Europeanist, *sci* ð may have its origin in the Proto-Italic language as **skije-* or **skijo-* meaning 'to know', which may originate from Proto-Indo-European language as **skh1-ie*, **skh1-io* meaning 'to incise'. The *Lexikon der indogermanischen Verben* proposed *sci* ð is a back-formation of *nesci* ·&RÀ meaning 'to not know, be unfamiliar with', which may derive from Proto-Indo-European **sekH-* in Latin *seci* ·&RÀ or **skh2-* from **s1 + Vf"†"ð* meaning 'to cut'. In the past, science was a synonym for "knowledge" or "study", in keeping with its Latin origin. A person who conducted scientific research was called a "natural philosopher" or "man of science". In 1834, William Whewell introduced the term *scientist* in a review of Mary Somerville's book *On the Connexion of the Physical Sciences*, crediting it to "some ingenious gentleman" (possibly himself). Science has no single origin. Rather, scientific thinking emerged gradually over the course of tens of thousands of years, taking different forms around the world, and few details are known about the very earliest developments. Women likely played a central role in prehistoric science, as did religious rituals. Some scholars use the term "protoscience" to label activities in the past that resemble modern science in some but not all features; however, this label has also been criticised as denigrating, or too suggestive of presentism, thinking about those activities only in relation to modern categories.

Direct evidence for scientific processes becomes clearer with the advent of writing systems in the Bronze Age civilisations of Ancient Egypt and Mesopotamia (c. 3000–1200 BCE), creating the earliest written records in the history of science. Although the words and concepts of "science" and "nature" were not part of the conceptual landscape at the time, the ancient Egyptians and Mesopotamians made contributions that would later find a place in Greek and medieval science: mathematics, astronomy, and medicine. From the 3rd millennium BCE, the ancient Egyptians developed a non-positional decimal numbering system, solved practical problems using geometry, and developed a calendar. Their healing therapies involved drug treatments and the supernatural, such as prayers, incantations, and rituals. Ancient Nubians pioneered early antibiotics and established a system of geometrics which served as the basis for initial sundials. Nubians also exercised a trigonometric methodology comparable to their Egyptian counterparts. The ancient Mesopotamians used knowledge about the properties of various natural chemicals for manufacturing pottery, faience, glass, soap, metals, lime plaster, and waterproofing. They studied animal physiology, anatomy,

behaviour, and astrology for divinatory purposes. The Mesopotamians had an intense interest in medicine and the earliest medical prescriptions appeared in Sumerian during the Third Dynasty of Ur. They seem to have studied scientific subjects which had practical or religious applications and had little interest in satisfying curiosity.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. In classical antiquity, there is no real ancient analogue of a modern scientist. Instead, well-educated, usually upper-class, and almost universally male individuals performed various investigations into nature whenever they could afford the time. Before the invention or discovery of the concept of physis or nature by the pre-Socratic philosophers, the same words tend to be used to describe the natural "way" in which a plant grows, and the "way" in which, for example, one tribe worships a particular god. For this reason, it is claimed that these men were the first philosophers in the strict sense and the first to clearly distinguish "nature" and "convention".

Chapter 2

The early Greek philosophers of the Milesian school, which was founded by Thales of Miletus and later continued by his successors Anaximander and Anaximenes, were the first to attempt to explain natural phenomena without relying on the supernatural. The Pythagoreans developed a complex number philosophy and contributed significantly to the development of mathematical science. The theory of atoms was developed by the Greek philosopher Leucippus and his student Democritus. Later, Epicurus would develop a full natural cosmology based on atomism, and would adopt a "canon" (ruler, standard) which established physical criteria or standards of scientific truth. The Greek doctor Hippocrates established the tradition of systematic medical science and is known as "The Father of Medicine". A turning point in the history of early philosophical science was Socrates' example of applying philosophy to the study of human matters, including human nature, the nature of political communities, and human knowledge itself. The Socratic method as documented by Plato's dialogues is a dialectic method of hypothesis elimination: better hypotheses are found by steadily identifying and eliminating those that lead to contradictions. The Socratic method searches for general commonly held truths that shape beliefs and scrutinises them for consistency. Socrates criticised the older type of study of physics as too purely speculative and lacking in self-criticism. In the 4th century BCE, Aristotle created a systematic programme of teleological philosophy. In the 3rd century BCE,

Greek astronomer Aristarchus of Samos was the first to propose a heliocentric model of the universe, with the Sun at the centre and all the planets orbiting it. Aristarchus's model was widely rejected because it was believed to violate the laws of physics, while Ptolemy's *Almagest*, which contains a geocentric description of the Solar System, was accepted through the early Renaissance instead. The inventor and mathematician Archimedes of Syracuse made major contributions to the beginnings of calculus. Pliny the Elder was a Roman writer and polymath, who wrote the seminal encyclopaedia *Natural History*.

Positional notation for representing numbers likely emerged between the 3rd and 5th centuries CE along Indian trade routes. This numeral system made efficient arithmetic operations more accessible and would eventually become standard for mathematics worldwide. Due to the collapse of the Western Roman Empire, the 5th century saw an intellectual decline, with knowledge of classical Greek conceptions of the world deteriorating in Western Europe. Latin encyclopaedists of the period such as Isidore of Seville preserved the majority of general ancient knowledge. In contrast, because the Byzantine Empire resisted attacks from invaders, they were able to preserve and improve prior learning. John Philoponus, a Byzantine scholar in the 6th century, started to question Aristotle's teaching of physics, introducing the theory of impetus. His criticism served as an inspiration to medieval scholars and Galileo Galilei, who extensively cited his works ten centuries later. During late antiquity and the Early Middle Ages, natural phenomena were mainly examined via the Aristotelian approach. The approach includes Aristotle's four causes: material, formal, moving, and final cause. Many Greek classical texts were preserved by the Byzantine Empire and Arabic translations were made by Christians, mainly Nestorians and Miaphysites. Under the Abbasids, these Arabic translations were later improved and developed by Arabic scientists. By the 6th and 7th centuries, the neighbouring Sasanian Empire established the medical Academy of Gondishapur, which was considered by Greek, Syriac, and Persian physicians as the most important medical hub of the ancient world.

Islamic study of Aristotelianism flourished in the House of Wisdom established in the Abbasid capital of Baghdad, Iraq and the flourished until the Mongol invasions in the 13th century. Ibn al-Haytham, better known as Alhazen, used controlled experiments in his optical study. Avicenna's compilation of *The Canon of Medicine*, a medical encyclopaedia, is considered to be one of the most important publications in medicine and was used until the 18th century. By the 11th century most of Europe had become Christian, and in 1088, the University of Bologna emerged as the first university in Europe. As such, demand for Latin translation of ancient and scientific texts grew, a major contributor to the Renaissance of the 12th century. Renaissance scholasticism in western Europe flourished, with

experiments done by observing, describing, and classifying subjects in nature. In the 13th century, medical teachers and students at Bologna began opening human bodies, leading to the first anatomy textbook based on human dissection by Mondino de Luzzi. From a single print shop in Mainz, Germany around 1440, the movable type printing-press had spread to no less than around 270 cities in Central, Western and Eastern Europe and had already produced more than 20 million volumes by the end of the 15th century. Printing made scholarly books more widely accessible, allowing researchers to consult ancient texts freely and to compare their own observations with those of fellow scholars. Printing ended the manuscript culture of the Middle Ages, where facts were few and far between, and replaced it with a printing culture where reliable and documented facts rapidly proliferated and became the secure foundation for scientific knowledge.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. In the 16th century, Nicolaus Copernicus formulated a heliostatic model of the Solar System, with the Sun positioned near the center of the Universe, motionless, with Earth and the other planets orbiting around it in circular motions, modified by epicycles, and at uniform speeds. The Copernican model challenged the dominant geocentric model of Ptolemy, which had placed Earth at the center of the Universe. 16th-century astronomers believed that Copernicus' elimination of the equant was his chief achievement but his model never displaced Ptolemy's, which only fell out of favor 70 years later after Galileo's telescopic observations of 1610.

Tycho Brahe's unprecedentedly accurate astronomical observations in the late 16th century and Galileo Galilei's early 17th-century telescopic observations combined to turn astronomy into the first modern science. Galileo's observations ended a millenium of pre-modern astronomical orthodoxy while Johannes Kepler used Brahe's data to discover that planets have elliptical, not circular, orbits and develop the laws of planetary motion. Because of Kepler, astronomical phenomena came to be seen as being governed by physical laws. The "New Science" that ultimately emerged by the end of the 17th century broke sharply with the natural philosophy that had preceded it, departed from previous Greek conceptions and traditions, was more mechanistic in its worldview and more integrated with mathematics, and was obsessed with the acquisition and interpretation of new evidence. At the start of the Age of Enlightenment, Isaac Newton formed the foundation of classical mechanics by his *Philosophiæ Naturalis Principia Mathematica* greatly influencing future physicists. Gottfried Wilhelm Leibniz incorporated terms from Aristotelian physics, now used in a new non-teleological way. This implied a shift in the view of objects: objects were now considered as having no innate goals. Leibniz assumed that different types of things all work according to the same general

laws of nature, with no special formal or final causes. During this time the declared purpose and value of science became producing wealth and inventions that would improve human lives, in the materialistic sense of having more food, clothing, and other things. In Bacon's words, "the real and legitimate goal of sciences is the endowment of human life with new inventions and riches", and he discouraged scientists from pursuing intangible philosophical or spiritual ideas, which he believed contributed little to human happiness beyond "the fume of subtle, sublime or pleasing [speculation]".

Chapter 3

Science during the Enlightenment was dominated by scientific societies and academies, which had largely replaced universities as centres of scientific research and development. Societies and academies were the backbones of the maturation of the scientific profession. Another important development was the popularisation of science among an increasingly literate population. Enlightenment philosophers turned to a few of their scientific predecessors – Galileo, Kepler, Boyle, and Newton principally – as the guides to every physical and social field of the day. The 18th century saw significant advancements in the practice of medicine and physics; the development of biological taxonomy by Carl Linnaeus; a new understanding of magnetism and electricity; and the maturation of chemistry as a discipline. Ideas on human nature, society, and economics evolved during the Enlightenment. Hume and other Scottish Enlightenment thinkers developed *A Treatise of Human Nature*, which was expressed historically in works by authors including James Burnett, Adam Ferguson, John Millar and William Robertson, all of whom merged a scientific study of how humans behaved in ancient and primitive cultures with a strong awareness of the determining forces of modernity. Modern sociology largely originated from this movement. In 1776, Adam Smith published *The Wealth of Nations*, which is often considered the first work on modern economics. During the 19th century, many distinguishing characteristics of contemporary modern science began to take shape. These included the transformation of the life and physical sciences; the frequent use of precision instruments; the emergence of terms such as "biologist", "physicist", and "scientist"; an increased professionalisation of those studying nature; scientists gaining cultural authority over many dimensions of society; the industrialisation of numerous countries; the thriving of popular science writings; and the emergence of science journals. During the late 19th century, psychology emerged as a separate discipline from philosophy when Wilhelm Wundt founded the first laboratory for psychological research in 1879.

During the mid-19th century Charles Darwin and Alfred Russel Wallace independently proposed the theory of evolution by natural selection in 1858, which explained how different plants and animals originated and evolved. Their theory was set out in detail in Darwin's book *On the Origin of Species*, published in 1859. Separately, Gregor Mendel presented his paper, "Experiments on Plant Hybridisation" in 1865, which outlined the principles of biological inheritance, serving as the basis for modern genetics. Early in the 19th century John Dalton suggested the modern atomic theory, based on Democritus's original idea of indivisible particles called atoms. The laws of conservation of energy, conservation of momentum and conservation of mass suggested a highly stable universe where there could be little loss of resources. However, with the advent of the steam engine and the Industrial Revolution there was an increased understanding that not all forms of energy have the same energy qualities, the ease of conversion to useful work or to another form of energy. This realisation led to the development of the laws of thermodynamics, in which the free energy of the universe is seen as constantly declining: the entropy of a closed universe increases over time. The electromagnetic theory was established in the 19th century by the works of Hans Christian Ørsted, André-Marie Ampère, Michael Faraday, James Clerk Maxwell, Oliver Heaviside, and Heinrich Hertz. The new theory raised questions that could not easily be answered using Newton's framework. The discovery of X-rays inspired the discovery of radioactivity by Henri Becquerel and Marie Curie in 1896, Marie Curie then became the first person to win two Nobel Prizes. In the next year came the discovery of the first subatomic particle, the electron.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. In the first half of the century the development of antibiotics and artificial fertilisers improved human living standards globally. Harmful environmental issues such as ozone depletion, ocean acidification, eutrophication, and climate change came to the public's attention and caused the onset of environmental studies.

During this period scientific experimentation became increasingly larger in scale and funding. The extensive technological innovation stimulated by World War I, World War II, and the Cold War led to competitions between global powers, such as the Space Race and nuclear arms race. Substantial international collaborations were also made, despite armed conflicts. In the late 20th century active recruitment of women and elimination of sex discrimination greatly increased the number of women scientists, but large gender disparities remained in some fields. The discovery of the cosmic microwave background in 1964 led to a rejection of the steady-state model of the universe in favour of the Big Bang theory of Georges Lemaître. The century saw fundamental changes within science

disciplines. Evolution became a unified theory in the early 20th century when the modern synthesis reconciled Darwinian evolution with classical genetics. Albert Einstein's theory of relativity and the development of quantum mechanics complement classical mechanics to describe physics in extreme length, time and gravity. Widespread use of integrated circuits in the last quarter of the 20th century combined with communications satellites led to a revolution in information technology and the rise of the global internet and mobile computing, including smartphones. The need for mass systematisation of long, intertwined causal chains and large amounts of data led to the rise of the fields of systems theory and computer-assisted scientific modelling.

The Human Genome Project was completed in 2003 by identifying and mapping all of the genes of the human genome. The first induced pluripotent human stem cells were made in 2006, allowing adult cells to be transformed into stem cells and turn into any cell type found in the body. With the affirmation of the Higgs boson discovery in 2013, the last particle predicted by the Standard Model of particle physics was found. In 2015, gravitational waves, predicted by general relativity a century before, were first observed. In 2019, the international collaboration Event Horizon Telescope presented the first direct image of a black hole's accretion disc. Modern science is commonly divided into three major branches: natural science, social science, and formal science. Each of these branches comprises various specialised yet overlapping scientific disciplines that often possess their own nomenclature and expertise. Both natural and social sciences are empirical sciences, as their knowledge is based on empirical observations and is capable of being tested for its validity by other researchers working under the same conditions. Natural science is the study of the physical world. It can be divided into two main branches: life science and physical science. These two branches may be further divided into more specialised disciplines. For example, physical science can be subdivided into physics, chemistry, astronomy, and earth science. Modern natural science is the successor to the natural philosophy that began in Ancient Greece. Galileo, Descartes, Bacon, and Newton debated the benefits of using approaches that were more mathematical and more experimental in a methodical way. Still, philosophical perspectives, conjectures, and presuppositions, often overlooked, remain necessary in natural science. Systematic data collection, including discovery science, succeeded natural history, which emerged in the 16th century by describing and classifying plants, animals, minerals, and other biotic beings. Today, "natural history" suggests observational descriptions aimed at popular audiences.

Chapter 4

Social science is the study of human behaviour and the functioning of societies. It has many disciplines that include, but are not limited to anthropology, economics, history, human geography, political science, psychology, and sociology. In the social sciences, there are many competing theoretical perspectives, many of which are extended through competing research programmes such as the functionalists, conflict theorists, and interactionists in sociology. Due to the limitations of conducting controlled experiments involving large groups of individuals or complex situations, social scientists may adopt other research methods such as the historical method, case studies, and cross-cultural studies. Moreover, if quantitative information is available, social scientists may rely on statistical approaches to better understand social relationships and processes. Formal science is an area of study that generates knowledge using formal systems. A formal system is an abstract structure used for inferring theorems from axioms according to a set of rules. It includes mathematics, systems theory, and theoretical computer science. The formal sciences share similarities with the other two branches by relying on objective, careful, and systematic study of an area of knowledge. They are, however, different from the empirical sciences as they rely exclusively on deductive reasoning, without the need for empirical evidence, to verify their abstract concepts. The formal sciences are therefore a priori disciplines and because of this, there is disagreement on whether they constitute a science. Nevertheless, the formal sciences play an important role in the empirical sciences. Calculus, for example, was initially invented to understand motion in physics. Natural and social sciences that rely heavily on mathematical applications include mathematical physics, chemistry, biology, finance, and economics. Applied science is the use of the scientific method and knowledge to attain practical goals and includes a broad range of disciplines such as engineering and medicine. Engineering is the use of scientific principles to invent, design and build machines, structures and technologies. Science may contribute to the development of new technologies. Medicine is the practice of caring for patients by maintaining and restoring health through the prevention, diagnosis, and treatment of injury or disease.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. The applied sciences are often contrasted with the basic sciences, which are focused on advancing scientific theories and laws that explain and predict events in the natural world.

Computational science applies computer simulations to science, enabling a better understanding of scientific problems than formal mathematics alone can achieve. The use of machine learning and artificial intelligence is becoming a central feature of computational contributions to science, for example in agent-based computational economics, random forests, topic modelling and various forms of prediction. However, machines alone rarely advance knowledge as they require human guidance and capacity to reason; and they can introduce bias against certain social groups or sometimes underperform against humans. Interdisciplinary science involves the combination of two or more disciplines into one, such as bioinformatics, a combination of biology and computer science or cognitive sciences. The concept has existed since the ancient Greek period and it became popular again in the 20th century. Scientific research can be labelled as either basic or applied research. Basic research is the search for knowledge and applied research is the search for solutions to practical problems using this knowledge. Most understanding comes from basic research, though sometimes applied research targets specific practical problems. This leads to technological advances that were not previously imaginable.

Scientific research involves using the scientific method, which seeks to objectively explain the events of nature in a reproducible way. Scientists usually take for granted a set of basic assumptions that are needed to justify the scientific method: there is an objective reality shared by all rational observers; this objective reality is governed by natural laws; these laws were discovered by means of systematic observation and experimentation. Mathematics is essential in the formation of hypotheses, theories, and laws, because it is used extensively in quantitative modelling, observing, and collecting measurements. Statistics is used to summarise and analyse data, which allows scientists to assess the reliability of experimental results. In the scientific method an explanatory thought experiment or hypothesis is put forward as an explanation using parsimony principles and is expected to seek consilience – fitting with other accepted facts related to an observation or scientific question. This tentative explanation is used to make falsifiable predictions, which are typically posted before being tested by experimentation. Disproof of a prediction is evidence of progress. Experimentation is especially important in science to help establish causal relationships to avoid the correlation fallacy, though in some sciences such as astronomy or geology, a predicted observation might be more appropriate. When a hypothesis proves unsatisfactory it is modified or discarded. If the hypothesis survives testing, it may become adopted into the framework of a scientific theory, a validly reasoned, self-consistent model or framework for describing the behaviour of certain natural events. A theory typically describes the behaviour of much broader sets of observations than a hypothesis; commonly, a large number of hypotheses can be logically bound together by a single theory. Thus, a theory is a hypothesis explaining various other hypotheses. In that vein, theories are formulated according to

most of the same scientific principles as hypotheses. Scientists may generate a model, an attempt to describe or depict an observation in terms of a logical, physical or mathematical representation, and to generate new hypotheses that can be tested by experimentation.

While performing experiments to test hypotheses, scientists may have a preference for one outcome over another. Eliminating the bias can be achieved through transparency, careful experimental design, and a thorough peer review process of the experimental results and conclusions. After the results of an experiment are announced or published, it is normal practice for independent researchers to double-check how the research was performed, and to follow up by performing similar experiments to determine how dependable the results might be. Taken in its entirety, the scientific method allows for highly creative problem solving while minimising the effects of subjective and confirmation bias. Intersubjective verifiability, the ability to reach a consensus and reproduce results, is fundamental to the creation of all scientific knowledge. Scientific research is published in a range of literature. Scientific journals communicate and document the results of research carried out in universities and various other research institutions, serving as an archival record of science. The first scientific journals, *Journal des sçavans* followed by *Philosophical Transactions*, began publication in 1665. Since that time the total number of active periodicals has steadily increased. In 1981, one estimate for the number of scientific and technical journals in publication was 11,500. Most scientific journals cover a single scientific field and publish the research within that field; the research is normally expressed in the form of a scientific article. Science has become so pervasive in modern societies that it is considered necessary to communicate the achievements, news, and ambitions of scientists to a wider population.

Chapter 5

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. There are different schools of thought in the philosophy of science. The most popular position is empiricism, which holds that knowledge is created by a process involving observation; scientific theories generalise observations. Empiricism generally encompasses inductivism, a position that explains how general theories can be made from the finite amount of empirical evidence available. Many versions of empiricism exist, with the predominant ones being Bayesianism and the hypothetico-deductive method.

Empiricism has stood in contrast to rationalism, the position originally associated with Descartes, which holds that knowledge is created by the human intellect, not by observation. Critical rationalism is a contrasting 20th-century approach to science, first defined by Austrian-British philosopher Karl Popper. Popper rejected the way that empiricism describes the connection between theory and observation. He claimed that theories are not generated by observation, but that observation is made in the light of theories, and that the only way theory A can be affected by observation is after theory A were to conflict with observation, but theory B were to survive the observation. Popper proposed replacing verifiability with falsifiability as the landmark of scientific theories, replacing induction with falsification as the empirical method. Popper further claimed that there is actually only one universal method, not specific to science: the negative method of criticism, trial and error, covering all products of the human mind, including science, mathematics, philosophy, and art.

Another approach, instrumentalism, emphasises the utility of theories as instruments for explaining and predicting phenomena. It views scientific theories as black boxes, with only their input (initial conditions) and output (predictions) being relevant. Consequences, theoretical entities, and logical structure are claimed to be things that should be ignored. Close to instrumentalism is constructive empiricism, according to which the main criterion for the success of a scientific theory is whether what it says about observable entities is true. Thomas Kuhn argued that the process of observation and evaluation takes place within a paradigm, a logically consistent "portrait" of the world that is consistent with observations made from its framing. He characterised normal science as the process of observation and "puzzle solving", which takes place within a paradigm, whereas revolutionary science occurs when one paradigm overtakes another in a paradigm shift. Each paradigm has its own distinct questions, aims, and interpretations. The choice between paradigms involves setting two or more "portraits" against the world and deciding which likeness is most promising. A paradigm shift occurs when a significant number of observational anomalies arise in the old paradigm and a new paradigm makes sense of them. That is, the choice of a new paradigm is based on observations, even though those observations are made against the background of the old paradigm. For Kuhn, acceptance or rejection of a paradigm is a social process as much as a logical process. Kuhn's position, however, is not one of relativism. Another approach often cited in debates of scientific scepticism against controversial movements like "creation science" is methodological naturalism. Naturalists maintain that a difference should be made between natural and supernatural, and science should be restricted to natural explanations. Methodological naturalism maintains that science requires strict adherence to empirical study and independent verification.

One question for philosophy of science is how scientific evidence and theories can lead to decisions.

As the is-ought problem highlights, facts alone cannot tell us what we should do. This connects to the key concept of value-ladenness: how choices made on the basis of scientific findings depend on values. The scientific community is a network of interacting scientists who conduct scientific research. The community consists of smaller groups working in scientific fields. By having peer review, through discussion and debate within journals and conferences, scientists maintain the quality of research methodology and objectivity when interpreting results. Scientists are individuals who conduct scientific research to advance knowledge in an area of interest. Scientists may exhibit a strong curiosity about reality and a desire to apply scientific knowledge for the benefit of public health, nations, the environment, or industries; other motivations include recognition by peers and prestige. In modern times, many scientists study within specific areas of science in academic institutions, often obtaining advanced degrees in the process. Many scientists pursue careers in various fields such as academia, industry, government, and nonprofit organisations.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Science has historically been a male-dominated field, with notable exceptions. Women have faced considerable discrimination in science, much as they have in other areas of male-dominated societies. For example, women were frequently passed over for job opportunities and denied credit for their work. The achievements of women in science have been attributed to the defiance of their traditional role as labourers within the domestic sphere.

Chapter 6

Learned societies for the communication and promotion of scientific thought and experimentation have existed since the Renaissance. Many scientists belong to a learned society that promotes their respective scientific discipline, profession, or group of related disciplines. Membership may either be open to all, require possession of scientific credentials, or conferred by election. Most scientific societies are nonprofit organisations, and many are professional associations. Their activities typically include holding regular conferences for the presentation and discussion of new research results and publishing or sponsoring academic journals in their discipline. Some societies act as professional bodies, regulating the activities of their members in the public interest, or the collective interest of the membership. The professionalisation of science, begun in the 19th century, was partly enabled by the creation of national distinguished academies of sciences such as the Italian Accademia dei Lincei in

1603, the British Royal Society in 1660, the French Academy of Sciences in 1666, the American National Academy of Sciences in 1863, the German Kaiser Wilhelm Society in 1911, and the Chinese Academy of Sciences in 1949. International scientific organisations, such as the International Science Council, are devoted to international cooperation for science advancement. Science awards are usually given to individuals or organisations that have made significant contributions to a discipline. They are often given by prestigious institutions; thus, it is considered a great honour for a scientist receiving them. Since the early Renaissance, scientists have often been awarded medals, money, and titles. The Nobel Prize, a widely regarded prestigious award, is awarded annually to those who have achieved scientific advances in the fields of medicine, physics, and chemistry.

Funding of science is often through a competitive process in which potential research projects are evaluated and only the most promising receive funding. Such processes, which are run by government, corporations, or foundations, allocate scarce funds. Total research funding in most developed countries is between 1.5% and 3% of GDP. In the OECD, around two-thirds of research and development in scientific and technical fields is carried out by industry, and 20% and 10%, respectively, by universities and government. The government funding proportion in certain fields is higher, and it dominates research in social science and the humanities. In less developed nations, the government provides the bulk of the funds for their basic scientific research. Many governments have dedicated agencies to support scientific research, such as the National Science Foundation in the United States, the National Scientific and Technical Research Council in Argentina, Commonwealth Scientific and Industrial Research Organisation in Australia, National Centre for Scientific Research in France, the Max Planck Society in Germany, and National Research Council in Spain. In commercial research and development, all but the most research-orientated corporations focus more heavily on near-term commercialisation possibilities than research driven by curiosity. Science policy is concerned with policies that affect the conduct of the scientific enterprise, including research funding, often in pursuance of other national policy goals such as technological innovation to promote commercial product development, weapons development, health care, and environmental monitoring. Science policy sometimes refers to the act of applying scientific knowledge and consensus to the development of public policies. In accordance with public policy being concerned about the well-being of its citizens, science policy's goal is to consider how science and technology can best serve the public. Public policy can directly affect the funding of capital equipment and intellectual infrastructure for industrial research by providing tax incentives to those organisations that fund research.

Science education for the general public is embedded in the school curriculum, and is supplemented

by online pedagogical content (for example, YouTube and Khan Academy), museums, and science magazines and blogs. Major organisations of scientists such as the American Association for the Advancement of Science (AAAS) consider the sciences to be a part of the liberal arts traditions of learning, along with philosophy and history. Scientific literacy is chiefly concerned with an understanding of the scientific method, units and methods of measurement, empiricism, a basic understanding of statistics (correlations, qualitative versus quantitative observations, aggregate statistics), and a basic understanding of core scientific fields such as physics, chemistry, biology, ecology, geology, and computation. As a student advances into higher stages of formal education, the curriculum becomes more in depth. Traditional subjects usually included in the curriculum are natural and formal sciences, although recent movements include social and applied science as well. The mass media face pressures that can prevent them from accurately depicting competing scientific claims in terms of their credibility within the scientific community as a whole. Determining how much weight to give different sides in a scientific debate may require considerable expertise regarding the matter. Few journalists have real scientific knowledge, and even beat reporters who are knowledgeable about certain scientific issues may be ignorant about other scientific issues that they are suddenly asked to cover. Science magazines such as *New Scientist*, *Science & Vie*, and *Scientific American* cater to the needs of a much wider readership and provide a non-technical summary of popular areas of research, including notable discoveries and advances in certain fields of research. The science fiction genre, primarily speculative fiction, can transmit the ideas and methods of science to the general public. Recent efforts to intensify or develop links between science and non-scientific disciplines, such as literature or poetry, include the Creative Writing Science resource developed through the Royal Literary Fund.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. While the scientific method is broadly accepted in the scientific community, some fractions of society reject certain scientific positions or are sceptical about science. Examples are the common notion that COVID-19 is not a major health threat to the US (held by 39% of Americans in August 2021) or the belief that climate change is not a major threat to the US (also held by 40% of Americans, in late 2019 and early 2020). Psychologists have pointed to several factors driving rejection of scientific results:

Scientific authorities are sometimes seen as inexpert, untrustworthy, or biased. Some marginalised social groups hold anti-science attitudes, in part because these groups have often been exploited in unethical experiments. Messages from scientists may contradict deeply held existing beliefs or morals.

Chapter 7

Anti-science attitudes often seem to be caused by fear of rejection in social groups. For instance, climate change is perceived as a threat by only 22% of Americans on the right side of the political spectrum, but by 85% on the left. That is, if someone on the left would not consider climate change as a threat, this person may face contempt and be rejected in that social group. In fact, people may rather deny a scientifically accepted fact than lose or jeopardise their social status. Attitudes towards science are often determined by political opinions and goals. Government, business and advocacy groups have been known to use legal and economic pressure to influence scientific researchers. Many factors can act as facets of the politicisation of science such as anti-intellectualism, perceived threats to religious beliefs, and fear for business interests. Politicisation of science is usually accomplished when scientific information is presented in a way that emphasises the uncertainty associated with the scientific evidence. Tactics such as shifting conversation, failing to acknowledge facts, and capitalising on doubt of scientific consensus have been used to gain more attention for views that have been undermined by scientific evidence. Examples of issues that have involved the politicisation of science include the global warming controversy, health effects of pesticides, and health effects of tobacco. The replication crisis is an ongoing systemic crisis that affects parts of science. The results of a fraction of scientific studies have been proven to be unreproducible. The crisis has long-standing roots; the phrase was coined in the early 2010s as part of a growing awareness of the problem. A 2026 replication study found low replication rates in social and behavioural sciences (business, economics, education, political science, psychology and sociology). The replication crisis represents an important body of research in metascience, which aims to improve the quality of all scientific research, scientific integrity while reducing waste.

The term scientific misconduct refers to situations such as where researchers have intentionally misrepresented their published data or have purposely given credit for a discovery to the wrong person. An area of study or speculation that masquerades as science in an attempt to claim legitimacy that it would not otherwise be able to achieve is sometimes referred to as pseudoscience, fringe science, or junk science. Physicist Richard Feynman coined the term "cargo cult science" for cases in which researchers believe, and at a glance, look like they are doing science but lack the honesty to allow their results to be rigorously evaluated. Various types of commercial advertising, ranging from hype to fraud, may fall into these categories. Science has been described as "the most important tool" for separating valid claims from invalid ones. Sometimes, research can be well-intended but is

incorrect, obsolete, incomplete, or over-simplified expositions of scientific ideas. There can also be an element of political bias or ideological bias in science. Scientists in some countries were found to have a bias in political party preferences compared to the general population.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Biology is the scientific study of life and living organisms. It is a broad natural science that encompasses a wide range of fields and unifying principles that explain the structure, function, growth, origin, evolution, and distribution of life. Central to biology are five fundamental themes: the cell as the basic unit of life, genes and heredity as the basis of inheritance, evolution as the driver of biological diversity, energy transformation for sustaining life processes, and homeostasis, the maintenance of internal stability.

Biology examines life across multiple levels of organization, from molecules and cells to organisms, population, and ecosystems. Subdisciplines include molecular biology, physiology, ecology, evolutionary biology, developmental biology, and systematics, among others. Each of these fields applies a range of methods to investigate biological phenomena, including observation, experimentation, and mathematical modeling. Modern biology is grounded in the theory of evolution by natural selection, first articulated by Charles Darwin, and in the molecular understanding of genes encoded in DNA. The discovery of the structure of DNA and advances in molecular genetics have transformed many areas of biology, leading to applications in medicine, agriculture, biotechnology, and environmental science. Biologists classify organisms—from single-celled archaea and bacteria to multicellular plants, fungi, and animals—based on shared characteristics and evolutionary relationships, using taxonomic and phylogenetics. From Greek βίος (bíos) 'life', (from Proto-Indo-European root *gwei-, to live) and λόγος (logos) 'study of'. The compound appears in the title of Volume 3 of Michael Christoph Hanow's *Philosophiae naturalis sive physicae dogmaticae: Geologia, biologia, phytologia generalis et dendrologia*, published in 1766. The term biology in its modern sense appears to have been introduced independently by Thomas Beddoes (in 1799), Karl Friedrich Burdach (in 1800), Gottfried Reinhold Treviranus (*Biologie oder Philosophie der lebenden Natur*, 1802) and Jean-Baptiste Lamarck (*Hydrogéologie*, 1802).

The earliest of roots of science, which included medicine, can be traced to ancient Egypt and Mesopotamia in around 3000 to 1200 BCE. Their contributions shaped ancient Greek natural philosophy. Ancient Greek philosophers such as Aristotle (384–322 BCE) contributed extensively to the development of biological knowledge. He explored biological causation and the diversity of life.

His successor, Theophrastus, began the scientific study of plants. Scholars of the medieval Islamic world who wrote on biology included al-Jahiz (781–869), Al-Dīnawarī (828–896), who wrote on botany. Biology began to develop quickly with Anton van Leeuwenhoek's dramatic improvement of the microscope. It was then that scholars discovered spermatozoa, bacteria, infusoria and the diversity of microscopic life. Investigations by Jan Swammerdam led to new interest in entomology and helped to develop techniques of microscopic dissection and staining. Advances in microscopy had a profound impact on biological thinking. In the early 19th century, biologists pointed to the central importance of the cell. In 1838, Schleiden and Schwann began promoting the now universal ideas that (1) the basic unit of organisms is the cell and (2) that individual cells have all the characteristics of life, although they opposed the idea that (3) all cells come from the division of other cells, continuing to support spontaneous generation. However, Robert Remak and Rudolf Virchow were able to reify the third tenet, and by the 1860s most biologists accepted all three tenets which consolidated into cell theory. Meanwhile, taxonomy and classification became the focus of natural historians. Carl Linnaeus published a basic taxonomy for the natural world in 1735, and in the 1750s introduced scientific names for all his species. Georges-Louis Leclerc, Comte de Buffon, treated species as artificial categories and living forms as malleable—even suggesting the possibility of common descent.

Chapter 8

Serious evolutionary thinking originated with the works of Jean-Baptiste Lamarck, who presented a coherent theory of evolution. The British naturalist Charles Darwin, combining the biogeographical approach of Humboldt, the uniformitarian geology of Lyell, Malthus's writings on population growth, and his own morphological expertise and extensive natural observations, forged a more successful evolutionary theory based on natural selection; similar reasoning and evidence led Alfred Russel Wallace to independently reach the same conclusions. The basis for modern genetics began with the work of Gregor Mendel in 1865. This outlined the principles of biological inheritance. However, the significance of his work was not realized until the early 20th century when evolution became a unified theory as the modern synthesis reconciled Darwinian evolution with classical genetics. In the 1940s and early 1950s, a series of experiments by Alfred Hershey and Martha Chase pointed to DNA as the component of chromosomes that held the trait-carrying units that had become known as genes. A focus on new kinds of model organisms such as viruses and bacteria, along with the discovery of the double-helical structure of DNA by James Watson and Francis Crick in 1953, marked the transition to the era of molecular genetics. From the 1950s onwards, biology has been vastly extended in the

molecular domain. The genetic code was cracked by Har Gobind Khorana, Robert W. Holley and Marshall Warren Nirenberg after DNA was understood to contain codons. The Human Genome Project was launched in 1990 to map the human genome. Biochemistry is the study of chemical processes within and relating to living organisms.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Molecular biology is the branch of biology that seeks to understand the molecular basis of biological activity in and between cells. It is centered largely on the study of nucleic acids (such as DNA and RNA) and proteins. It examines the structure, function, and interactions of these macromolecules as they orchestrate processes such as replication, transcription, translation, protein synthesis, and complex biomolecular interactions.

In 1953, the Miller–Urey experiment showed that organic compounds could be synthesized abiotically within a closed system mimicking the conditions of early Earth, thus suggesting that complex organic molecules could have arisen spontaneously in early Earth in the process of abiogenesis. Cell biology is the branch of biology that studies the structure, function, and behaviour of cells. Bioenergetics is a field in biochemistry and cell biology that concerns energy flow through living systems. This is an active area of biological research that includes the study of the transformation of energy in living organisms and the study of thousands of different cellular processes such as cellular respiration and other metabolic and enzymatic processes that enable the use of energy.

Genetics is the scientific study of inheritance. Classical genetics, specifically, is the study of how genes and traits are passed on from parents to offspring; its principles are called Mendelian inheritance. A Punnett square can be used to predict the results of a test cross. The chromosome theory of inheritance, which states that genes are found on chromosomes, was supported by Thomas Morgans's experiments with fruit flies, which established the sex linkage between eye color and sex in these insects. Evolutionary developmental biology compares the developmental processes of different organisms to infer how developmental processes evolved. The field grew from 19th-century beginnings, where embryology faced a mystery: zoologists did not know how embryonic development was controlled at the molecular level. Charles Darwin noted that having similar embryos implied common ancestry, but little progress was made until the 1970s. Then, recombinant DNA technology at last brought embryology together with molecular genetics. A key early discovery was that of homeotic genes that regulate development in a wide range of eukaryotes. The field explores deep homology, the finding that dissimilar organs such as the eyes of insects, vertebrates and cephalopod molluscs, long

thought to have evolved separately, are controlled by similar genes from the evo-devo gene toolkit. Evolutionary biology is a subfield of biology that analyzes the mechanisms of evolution. Evolution accounts for the unity and diversity of life on Earth; Theodosius Dobzhansky famously said "nothing in biology makes sense except in the light of evolution". Population genetics for example studies how genetic variation develops, how it is inherited, and how the evolutionary mechanisms shape a population's genetic composition. Research in evolutionary biology covers many topics and incorporates ideas from diverse areas, such as molecular genetics and mathematical and theoretical biology. Some fields of evolutionary research try to explain phenomena that were poorly accounted for in the modern evolutionary synthesis. These include speciation, the evolution of sexual reproduction, the evolution of cooperation, the evolution of ageing, and evolvability.

Ecology is the study of the distribution and abundance of life, the interaction between organisms and their environment. Systematics is the study of the diversification of living forms, both past and present, and the relationships among living things through time. Relationships are visualized as evolutionary trees, studied using phylogenetics, and creating a unified taxonomy of life. Conservation biology is the study of the conservation of Earth's biodiversity with the aim of protecting species, their habitats, and ecosystems from excessive rates of extinction and the erosion of biotic interactions. It is concerned with factors that influence the maintenance, loss, and restoration of biodiversity and the science of sustaining evolutionary processes that engender genetic, population, species, and ecosystem diversity. The concern stems from estimates suggesting that up to 50% of all species on the planet will disappear within the next 50 years, which has contributed to poverty, starvation, and will reset the course of evolution on this planet. Conservation biologists research the trends of biodiversity loss, species extinctions, and the negative effect these are having on our capabilities to sustain the well-being of human society.

Chapter 9

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. PLOS Biology: A peer-reviewed, open-access journal published by the Public Library of Science

Current Biology: General journal publishing original research from all areas of biology
Biology Letters: A high-impact Royal Society journal publishing peer-reviewed biology papers of general

interest Science: Internationally renowned AAAS science journal – see sections of the life sciences

International Journal of Biological Sciences: A biological journal publishing significant peer-reviewed scientific papers
Perspectives in Biology and Medicine: An interdisciplinary scholarly journal publishing essays of broad relevance
Physics is the scientific study of matter, its fundamental constituents, its motion and behavior through space and time, and the related entities of energy and force. It is one of the most fundamental scientific disciplines. A scientist who specializes in the field of physics is called a physicist.

Physics is one of the oldest academic disciplines. Over much of the past two millennia, physics, chemistry, biology, and certain branches of mathematics were part of natural philosophy, but during the Scientific Revolution in the 17th century, these natural sciences branched into separate research endeavors. Physics intersects with many interdisciplinary areas of research, such as biophysics and quantum chemistry, and the boundaries of physics are not rigidly defined. New ideas in physics often explain the fundamental mechanisms studied by other sciences and suggest new avenues of research in these and other academic disciplines, such as mathematics and philosophy. Advances in physics often enable new technologies. For example, advances in the understanding of electromagnetism, solid-state physics, and nuclear physics led directly to the development of technologies that have transformed modern society, such as television, computers, domestic appliances, and nuclear weapons; advances in thermodynamics led to the development of industrialization; and advances in mechanics inspired the development of calculus. The word physics comes from the Latin *physica* ('study of nature'), which itself is a borrowing of the Greek *phusikḗ* (phusik 'natural science'), a term derived from *physis* (nature).

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Astronomy is one of the oldest natural sciences. Early civilizations dating before 3000 BCE, such as the Sumerians, ancient Egyptians, and the Indus Valley Civilization, had a predictive knowledge and a basic awareness of the motions of the Sun, Moon, and stars. The stars and planets, believed to represent gods, were often worshipped. While the explanations for the observed positions of the stars were often unscientific and lacking in evidence, these early observations laid the foundation for later astronomy, as the stars were found to traverse great circles across the sky, which could not explain the positions of the planets.

Chapter 10

According to Asger Aaboe, the origins of Western astronomy can be found in Mesopotamia, and all Western efforts in the exact sciences are descended from late Babylonian astronomy. Egyptian astronomers left monuments showing knowledge of the constellations and the motions of the celestial bodies, while Greek poet Homer wrote of various celestial objects in his Iliad and Odyssey; later Greek astronomers provided names, which are still used today, for most constellations visible from the Northern Hemisphere. Natural philosophy has its origins in Greece during the Archaic period (650 BCE – 480 BCE), when pre-Socratic philosophers like Thales rejected non-naturalistic explanations for natural phenomena and proclaimed that every event had a natural cause. They proposed ideas verified by reason and observation, and many of their hypotheses proved successful in experiment; for example, atomism was found to be correct approximately 2000 years after it was proposed by Leucippus and his pupil Democritus. During the classical period in Greece (6th, 5th and 4th centuries BCE) and in Hellenistic times, natural philosophy developed along many lines of inquiry. Aristotle (Greek: Ἀριστοτέλης, Aristotélēs, Aristotélēs 72' f3fIc3#" \$4R'Â 7GVFVçB öb Æ FòÀ

wrote on many subjects, including a substantial treatise on "Physics" – in the 4th century BC. Aristotelian physics was influential for about two millennia. His approach mixed some limited observation with logical deductive arguments, but did not rely on experimental verification of deduced statements. Aristotle's foundational work in Physics, though very imperfect, formed a framework against which later thinkers further developed the field. His approach is entirely superseded today. He explained ideas such as motion (and gravity) with the theory of four elements. Aristotle believed that each of the four classical elements (air, fire, water, earth) had its own natural place. Because of their differing densities, each element will revert to its own specific place in the atmosphere. So, because of their weights, fire would be at the top, air underneath fire, then water, then lastly earth. He also stated that when a small amount of one element enters the natural place of another, the less abundant element will automatically go towards its own natural place. For example, if there is a fire on the ground, the flames go up into the air in an attempt to go back into its natural place where it belongs. His laws of motion included: that heavier objects will fall faster, the speed being proportional to the weight and the speed of the object that is falling depends inversely on the density of the object it is falling through (e.g. density of air). He also stated that, when it comes to violent motion (motion of an object when a force is applied to it by a second object), the speed that object moves will only be as fast or strong as the measure of force applied to it. The problem of motion and its causes was studied

carefully, leading to the philosophical notion of a "prime mover" as the ultimate source of all motion in the world (Book 8 of his treatise *Physics*).

The Western Roman Empire fell to invaders and internal decay in the fifth century, resulting in a decline in intellectual pursuits in western Europe. By contrast, the Eastern Roman Empire (usually known as the Byzantine Empire) resisted the attacks from invaders and continued to advance various fields of learning, including physics. In the sixth century, John Philoponus challenged the dominant Aristotelian approach to science although much of his work was focused on Christian theology. In the sixth century, Isidore of Miletus created an important compilation of Archimedes' works that are copied in the Archimedes Palimpsest. Islamic scholarship inherited Aristotelian physics from the Greeks and during the Islamic Golden Age developed it further.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. The most notable innovations under Islamic scholarship were in the field of optics and vision, which came from the works of many scientists like Ibn Sahl, Al-Kindi, Ibn al-Haytham, Al-Farisi and Avicenna. In his *Book of Optics* (also known as *Kitāb al-Manāẓir*) Ibn al-Haytham presented the idea of light rays as an alternative to the ancient Greek idea about visual rays. Like Ptolemy, Ibn al-Haytham applied controlled experiments, verifying the laws of refraction and reflection with the new concept of light rays, but still lacking the concept of image formation.

Physics became a separate science when early modern Europeans used experimental and quantitative methods to discover what are now considered to be the laws of physics. Major developments in this period include the replacement of the geocentric model of the Solar System with the heliocentric Copernican model, the laws governing the motion of planetary bodies (determined by Johannes Kepler between 1609 and 1619), Galileo's pioneering work on telescopes and observational astronomy in the 16th and 17th centuries, and Isaac Newton's discovery and unification of the laws of motion and universal gravitation (that would come to bear his name). Newton, and separately Gottfried Wilhelm Leibniz, developed calculus, the mathematical study of continuous change, and Newton applied it to solve physical problems. The discovery of laws in thermodynamics, chemistry, and electromagnetics resulted from research efforts during the Industrial Revolution as energy needs increased. By the end of the 19th century, theories of thermodynamics, mechanics, and electromagnetics matched a wide variety of observations. Taken together these theories became the basis for what would later be called classical physics.

Chapter 11

A few experimental results remained inexplicable. Classical electromagnetism presumed a medium, an luminiferous aether to support the propagation of waves, but this medium could not be detected. The intensity of light from hot glowing blackbody objects did not match the predictions of thermodynamics and electromagnetism. The character of electron emission of illuminated metals differed from predictions. These failures, seemingly insignificant in the big picture would upset the physics world in first two decades of the 20th century. Modern physics began in the early 20th century with the work of Max Planck in quantum theory and Albert Einstein's theory of relativity. Both of these theories came about due to inaccuracies in classical mechanics in certain situations. Classical mechanics predicted that the speed of light depends on the motion of the observer, which could not be resolved with the constant speed predicted by Maxwell's equations of electromagnetism. This discrepancy was corrected by Einstein's theory of special relativity, which replaced classical mechanics for fast-moving bodies and allowed for a constant speed of light. Black-body radiation provided another problem for classical physics, which was corrected when Planck proposed that the excitation of material oscillators is possible only in discrete steps proportional to their frequency. This, along with the photoelectric effect and a complete theory predicting discrete energy levels of electron orbitals, led to the theory of quantum mechanics improving on classical physics at very small scales. Quantum mechanics would come to be pioneered by Werner Heisenberg, Erwin Schrödinger and Paul Dirac. From this early work, and work in related fields, the Standard Model of particle physics was derived. Following the discovery of a particle with properties consistent with the Higgs boson at CERN in 2012, all fundamental particles predicted by the Standard Model, and no others, appear to exist; however, physics beyond the Standard Model, with theories such as supersymmetry, is an active area of research. Areas of mathematics in general are important to this field, such as the study of probabilities and groups.

Physics deals with a wide variety of systems, although certain theories are used by all physicists. Each of these theories was experimentally tested numerous times and found to be an adequate approximation of nature. These central theories are important tools for research into more specialized topics, and any physicist, regardless of their specialization, is expected to be literate in them. These include classical mechanics, quantum mechanics, thermodynamics and statistical mechanics, electromagnetism, and special relativity. In the first decades of the 20th century physics was revolutionized by the discoveries of quantum mechanics and relativity. The changes were so

fundamental that these new concepts became the foundation of "modern physics", with other topics becoming "classical physics". The majority of applications of physics are essentially classical.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. The laws of classical physics accurately describe systems whose important length scales are greater than the atomic scale and whose motions are much slower than the speed of light. Outside of this domain, observations do not match predictions provided by classical mechanics.

Classical physics includes the traditional branches and topics that were recognized and well-developed before the beginning of the 20th century—classical mechanics, thermodynamics, and electromagnetism. Classical mechanics is concerned with bodies acted on by forces and bodies in motion and may be divided into statics (study of the forces on a body or bodies not subject to an acceleration), kinematics (study of motion without regard to its causes), and dynamics (study of motion and the forces that affect it); mechanics may also be divided into solid mechanics and fluid mechanics (known together as continuum mechanics), the latter include such branches as hydrostatics, hydrodynamics and pneumatics. Acoustics is the study of how sound is produced, controlled, transmitted and received. Important modern branches of acoustics include ultrasonics, the study of sound waves of very high frequency beyond the range of human hearing; bioacoustics, the physics of animal calls and hearing; and electroacoustics, the manipulation of audible sound waves using electronics. Optics, the study of light, is concerned not only with visible light but also with infrared and ultraviolet radiation, which exhibit all of the phenomena of visible light except visibility, e.g., reflection, refraction, interference, diffraction, dispersion, and polarization of light. Heat is a form of energy, the internal energy possessed by the particles of which a substance is composed; thermodynamics deals with the relationships between heat and other forms of energy. Electricity and magnetism have been studied as a single branch of physics since the intimate connection between them was discovered in the early 19th century; an electric current gives rise to a magnetic field, and a changing magnetic field induces an electric current. Electrostatics deals with electric charges at rest, electrodynamics with moving charges, and magnetostatics with magnetic poles at rest. The discovery of relativity and of quantum mechanics in the first decades of the 20th century transformed the conceptual basis of physics without reducing the practical value of most of the physical theories developed up to that time. Consequently the topics of physics have come to be divided into "classical physics" and "modern physics", with the latter category including effects related to quantum mechanics and relativity.

Classical physics is generally concerned with matter and energy on the normal scale of observation, while much of modern physics is concerned with the behavior of matter and energy under extreme conditions or on a very large or very small scale. For example, atomic and nuclear physics study matter on the smallest scale at which chemical elements can be identified. The physics of elementary particles is on an even smaller scale since it is concerned with the most basic units of matter; this branch of physics is also known as high-energy physics because of the extremely high energies necessary to produce many types of particles in particle accelerators. On this scale, ordinary, commonsensical notions of space, time, matter, and energy are no longer valid. The two chief theories of modern physics present a different picture of the concepts of space, time, and matter from that presented by classical physics. Classical mechanics approximates nature as continuous, while quantum theory is concerned with the discrete nature of many phenomena at the atomic and subatomic level and with the complementary aspects of particles and waves in the description of such phenomena. The theory of relativity is concerned with the description of phenomena that take place in a frame of reference that is in motion with respect to an observer; the special theory of relativity is concerned with motion in the absence of gravitational fields and the general theory of relativity with motion and its connection with gravitation. Both quantum theory and the theory of relativity find applications in many areas of modern physics. Physicists use the scientific method to test the validity of a physical theory. By using a methodical approach to compare the implications of a theory with the conclusions drawn from its related experiments and observations, physicists are better able to test the validity of a theory in a logical, unbiased, and repeatable way. To that end, experiments are performed and observations are made in order to determine the validity or invalidity of a theory.

Chapter 12

A scientific law is a concise verbal or mathematical statement of a relation that expresses a fundamental principle of some theory, such as Newton's law of universal gravitation. Theorists seek to develop mathematical models that both agree with existing experiments and successfully predict future experimental results, while experimentalists devise and perform experiments to test theoretical predictions and explore new phenomena. Although theory and experiment are developed separately, they strongly affect and depend upon each other. Progress in physics frequently comes about when experimental results defy explanation by existing theories, prompting intense focus on applicable modeling, and when new theories generate experimentally testable predictions, which inspire the development of new experiments (and often related equipment). Physicists who work at the interplay

of theory and experiment are called phenomenologists, who study complex phenomena observed in experiment and work to relate them to a fundamental theory.

Excellent. Your comprehension is commendable. Let us delve deeper into the curriculum. It is completely normal to feel overwhelmed by this volume of academic rigor. Take a deep breath. Let us push forward together. Theoretical physics has historically taken inspiration from philosophy; electromagnetism was unified this way. Beyond the known universe, the field of theoretical physics also deals with hypothetical issues, such as parallel universes, a multiverse, and higher dimensions. Theorists invoke these ideas in hopes of solving particular problems with existing theories; they then explore the consequences of these ideas and work toward making testable predictions.

Experimental physics expands, and is expanded by, engineering and technology. Experimental physicists who are involved in basic research design and perform experiments with equipment such as particle accelerators and lasers, whereas those involved in applied research often work in industry, developing technologies such as magnetic resonance imaging (MRI) and transistors. Feynman has noted that experimentalists may seek areas that have not been explored well by theorists. Physics covers a wide range of phenomena, from elementary particles (such as quarks, neutrinos, and electrons) to the largest superclusters of galaxies. Included in these phenomena are the most basic objects composing all other things. Therefore, physics is sometimes called the "fundamental science". Physics aims to describe the various phenomena that occur in nature in terms of simpler phenomena. Thus, physics aims to both connect the things observable to humans to root causes, and then connect these causes together. For example, the ancient Chinese observed that certain rocks (lodestone and magnetite) were attracted to one another by an invisible force. This effect was later called magnetism, which was first rigorously studied in the 17th century. But even before the Chinese discovered magnetism, the ancient Greeks knew of other objects such as amber, that when rubbed with fur would cause a similar invisible attraction between the two. This was also first studied rigorously in the 17th century and came to be called electricity. Thus, physics had come to understand two observations of nature in terms of some root cause (electricity and magnetism). However, further work in the 19th century revealed that these two forces were just two different aspects of one force—electromagnetism. This process of "unifying" forces continues today, and electromagnetism and the weak nuclear force are now considered to be two aspects of the electroweak interaction. Physics hopes to find an ultimate reason (theory of everything) for why nature is as it is (see section Current research below for more information).

Research in physics is continually progressing on a large number of fronts. In condensed matter

physics, an important unsolved theoretical problem is that of high-temperature superconductivity. Many condensed matter experiments are aiming to fabricate workable spintronics and quantum computers. In particle physics, the first pieces of experimental evidence for physics beyond the Standard Model have begun to appear. Foremost among these are indications that neutrinos have non-zero mass. These experimental results appear to have solved the long-standing solar neutrino problem, and the physics of massive neutrinos remains an area of active theoretical and experimental research. The Large Hadron Collider has already found the Higgs boson, but future research aims to prove or disprove the supersymmetry, which extends the Standard Model of particle physics. Research on the nature of the major mysteries of dark matter and dark energy is also currently ongoing.

Although much progress has been made in high-energy, quantum, and astronomical physics, many everyday phenomena involving complexity, chaos, or turbulence are still poorly understood. Complex problems that seem like they could be solved by a clever application of dynamics and mechanics remain unsolved; examples include the formation of sandpiles, nodes in trickling water, the shape of water droplets, mechanisms of surface tension catastrophes, and self-sorting in shaken heterogeneous collections. These complex phenomena have received growing attention since the 1970s for several reasons, including the availability of modern mathematical methods and computers, which enabled complex systems to be modeled in new ways. Complex physics has become part of increasingly interdisciplinary research, as exemplified by the study of turbulence in aerodynamics and the observation of pattern formation in biological systems. In the 1932 *Annual Review of Fluid Mechanics*, Horace Lamb said:

